VENKATESH SHANMUGAM

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LinkedIn | GitHub | Portfolio

SUMMARY

Senior AI Engineer with 5+ years building **production AI systems** for regulated enterprise GTM. At Aithon AI, joined as the **first AI engineer** and owned the **AI/ML layer** for a sales-intelligence platform fusing 10+ source families across **501K+ companies** into customer-ready plays. Built the **LLM gateway** behind **42.7M calls** across **17 services** in 30 days, cut LLM spend by **~70%**, and moved document extraction from **AWS Textract** to **open-source models** on **async SageMaker**, retiring about **$7.8K/month** in run-rate while owning **architecture, evals, governance, and AWS delivery**.

TECHNICAL SKILLS

AI / ML Systems: Production LLMs, agentic AI, LangGraph, RAG, GraphRAG, knowledge graphs, entity resolution, model evaluation, LoRA fine-tuning, PyTorch, Transformers
LLM Platforms & Evals: Provider routing, prompt and response caching, tool use, context engineering, batch inference, tracing, cost attribution, guardrails, security testing
Data & Retrieval: Python, TypeScript, SQL, Spark, Airflow, Databricks, Delta Lake, PostgreSQL, Neo4j, Elasticsearch, FastAPI, A/B testing, data-quality gates
Cloud & ML Infrastructure: AWS SageMaker async inference, S3, ECS, Step Functions, Glue, Athena, Terraform, Docker, Kubernetes, MLflow, CI/CD, observability
Technical Leadership: 0-to-1 platform architecture, distributed systems design, customer discovery, B2B GTM systems, production reliability, incident response

PROFESSIONAL EXPERIENCE

AITHON AI | *Senior AI Engineer* Oct 2025 – Present
New York, NY

- Owned **0-to-1 AI architecture** for Aithon's B2B GTM intelligence platform as **first AI engineer**, turning **10+ signal families** across **501K+ companies** into **10M+ scored lead, stakeholder, intent, and market-intelligence records** on a **production AWS service fleet**.
- Designed **evidence-first agentic company intelligence** with **RAG**, **graph reasoning**, **entity resolution**, citations, fallback behavior, and **evaluation gates**; preserved exact score equivalence in an **855-case validation** while connecting **millions of signals** into customer-ready narratives.
- Built the shared **Python and TypeScript Aithon LLM gateway** for **four providers** with routing, retries, **prompt and response caching**, **tracing**, **batch execution**, and **governance**; production ledger recorded **42.7M calls** across **17 services** in 30 days with **zero unpriced calls**.
- Moved document extraction from **AWS Textract** to **open-source GPU extraction models** served through **async SageMaker**; retired about **$7.8K/month** in Textract run-rate and improved benchmark table-cell capture to **82.3%**.
- Built an **agentic CRM miner** for **schema-drifting Salesforce and HubSpot fields**, extracting existing customer footprint into a **self-evolving account-memory layer** that grounds new intelligence; improved email-to-opportunity association from **0.74% to 12% (16x)** and cut peak miner memory from **~16GB to ~1.25GB**.

ScriptChain Health | *Machine Learning Engineer Intern* Jun 2025 – Sep 2025
Washington, DC

- Built a **Graph Attention Network** plus **time-aware LLM** readmission model for **30-day hospital prediction**, achieving **78% ROC-AUC** in clinical trials on an AI-powered metabolic syndrome platform.
- Architected a **LangGraph** and **NVIDIA NeMo Guardrails** platform for personalized nutrition workflows, combining **multi-agent orchestration** with **HIPAA-compliant FHIR data synchronization**.

Tata Consultancy Services | *Digital Transformation Developer* Apr 2021 – Aug 2023
Chennai, India

- Designed and deployed **12+ RPA bots** across **14+ client teams**, reducing infrastructure costs by **40%** and process turnaround by **30%** across enterprise operations.
- Integrated automation with **CRM, billing, and network-management systems** and standardized reusable delivery patterns spanning **3 business domains** and **14+ client teams**.

PROJECTS

Agentic Graph-RAG for Building Codes | Built production-scale multi-agent RAG over Virginia building codes with **5 LangGraph agents**, **Neo4j knowledge graph**, **Redis retrieval**, **LangSmith evals**, **90%** query-accuracy improvement, **40%** cost reduction, and **99.9% uptime**
AI-Text Discriminator | LoRA fine-tuned transformer on 1.2M+ text samples: 97% accuracy, **3x faster training**, **50% lower memory**, and **90% parameter reduction**

EDUCATION

The George Washington University, M.S. Computer Science, GPA: 3.88 / 4.0 Sep 2023 – May 2025
SRM University, B.Tech Computer Science, GPA: 3.5 / 4.0 Aug 2016 – May 2020